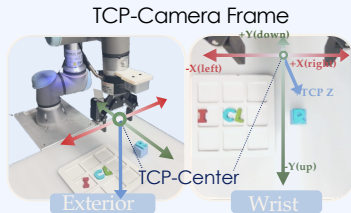
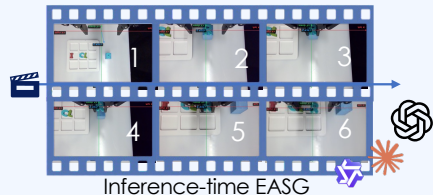


EASG

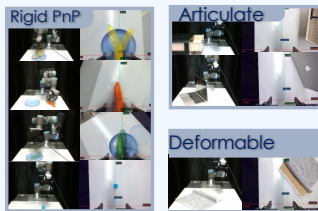


VLM-Native Action

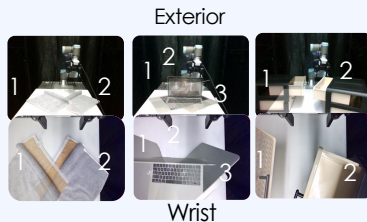
“Task: Pick “R” → target grid
 Format: taskstate | distance_cm | ...
 Rules: Set task_state to 1...
 Example: 0|-0.10,+0.91,+0.40,...”



✓ Zero-shot



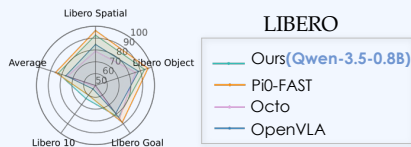
Zero-shot Task Type



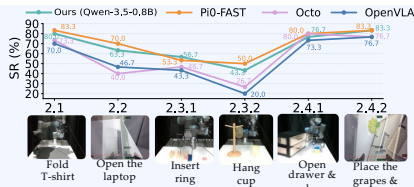
FREE Movement and
 Rotation
 within Manipulation Area

Randomization

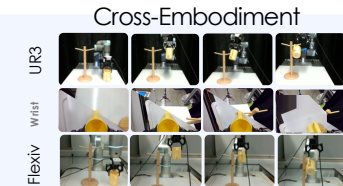
✓ Trainable



Simulation



Real-world



Trained w/ Wrist Obs & eef pose in TCP frame
 (Exterior Obs only for reference)